A Reference-Centered Clock Positivity Condition

A conditional next step for the six-lane receiving calculation

SAM Volume I research note • Sean Brady
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Result. Subtracting the stationary reference from the exact correlated clock equation yields a direct sufficient condition for instantaneous positivity. In the source regime $0 < \Omega \le \Delta$, it is enough to certify $2|g|W + \Delta E < \Omega M$, where E bounds the complex-current error and W bounds the signed joint-correlator error relative to that reference. The condition preserves the reference cancellation and defines a concrete accuracy target. It does not itself supply the missing correlator estimate or improve any native rate interval. Field-domain, weighted-moment and reference-state requirements are stated explicitly.

1. The exact centered identity

All quantities use the same normalized model-time chart. Retain the supplied source identity

$$\dot z = -i\Delta z + 2gw, \quad w = \langle S(\phi_\times - i\phi_+)\rangle, \tag{1}$$

with real $g \neq 0$, $\Delta > 0$, and the actual phase rate $\nu = -\operatorname{Im}(\dot z/z)$ where $z \neq 0$. The reference gap Ω is one fixed true parameter throughout the calculation. Let

$$h = Me^{-i\Omega t}, \quad w_h = \frac{i(\Delta-\Omega)h}{2g}, \quad M > 0. \tag{2}$$

Set $\delta z = z - h$ and $\delta w = w - w_h$. Because $\dot h = -i\Omega h$, direct subtraction gives

$$r := \dot z + i\Omega z = -i(\Delta-\Omega)\delta z + 2g\delta w. \tag{3}$$

Consequently

$$\boxed{\nu - \Omega = (\Delta-\Omega)\operatorname{Re}\frac{\delta z}{z} - 2g\operatorname{Im}\frac{\delta w}{z}.} \tag{4}$$

The sign of g remains in the exact identity. Using independent enclosures for w and w_h can discard their stationary cancellation. A direct enclosure of δw, or of r, is the more informative target.

2. A conditional positivity theorem

Suppose that at the time being tested

$$|\delta z| \le E < M, \qquad |\delta w| \le W. \tag{5}$$

Then $|z| \ge M - E > 0$. Equation (4) implies

$$\nu \ge \Omega - \frac{|\Delta-\Omega|E + 2|g|W}{M-E}. \tag{6}$$

For $\Omega > 0$, a sufficient condition for strict positivity is

$$|\Delta-\Omega|E + 2|g|W < \Omega(M-E). \tag{7}$$

Equality certifies at most nonnegativity. Failure of this sufficient test is inconclusive; it does not establish a nonpositive physical clock.

The recovered six-lane gap intervals satisfy $0 < \Omega < \Delta$. In that regime, the same true Ω on both sides cancels algebraically to give

$$\boxed{2|g|W + \Delta E < \Omega M.} \tag{8}$$

Equivalently, the target correlator accuracy is

$$W < W_{\text{target}} := \frac{\Omega M - \Delta E}{2|g|}, \tag{9}$$

provided its numerator is positive. This simplification must occur before unnecessary interval decorrelation is introduced.

The one-sided estimate is sharp given only the two separate disk bounds in this parameter regime. Rotate the phase frame so that $h = M$, and take $\delta z = -E$ and $\delta w = i\operatorname{sign}(g)W$. Equation (6) is then saturated. A uniformly stronger bound needs more information about correlation, phase geometry or dynamics.

3. Interval-safe implementation

The admissible set must retain a common Ω and the known relationships among g, $E(t)$, radius, split and preparation. A joint enclosure of

$$F = \Omega M - \Delta E - 2|g|W \tag{10}$$

with strictly positive lower endpoint proves the desired result. For a conservative rectangular superset, use

$$B(t) = \frac{\Omega_{\text{lo}} M_{\text{lo}} - \Delta E_{\text{up}}(t)}{2G_{\text{up}}}, \qquad G_{\text{up}} \ge |g|. \tag{11}$$

If $E_{\text{up}} < M_{\text{lo}}$, $B > 0$ and a valid correlator certificate gives $W_{\text{up}} < B$, positivity follows. Compute lower budgets with directed rounding or exact rational arithmetic. A printed rounded value is usable only if it is a rigorous lower bound for the budget.

A direct residual estimate can be better: $|r| \le R_{\text{err}} < \Omega(M - E)$ also proves positivity and can preserve cancellations lost by separate absolute values. Alternatively, evaluate the signed expression (4) directly over a common admissible set. No one of these tests is obtained by differentiating an accumulated-phase envelope. Endpoint certificates prove endpoint positivity; a full time interval requires uniform control.

4. What the missing field estimate requires

The quantity w_h is initially an algebraic reference. A state-comparison proof needs an actual reference state whose field expectation equals w_h, or a certified mismatch term. An exact hard stationary preparation with the soft vacuum is a natural candidate, but its identification must be established in the same field realization and domain.

Write $B_f = S(\phi_\times - i\phi_+)$. The boundedness of S does not make the field insertion bounded. There are at least three valid routes to a comparison estimate, each with substantive hypotheses.

4.1 Weighted form control

Let $A \geq I$ be a positive weight and suppose the field form has a bounded weighted representation

$$\|A^{-1/2} B_f A^{-1/2}\| \leq C_B. \tag{12}$$

For a well-defined weighted trace-class difference,

$$|\operatorname{Tr}[(\rho-\sigma)B_f]| \leq C_B \|A^{1/2}(\rho-\sigma)A^{1/2}\|_1. \tag{13}$$

Thus a certified weighted state-distance estimate, its form constant, and a reference identification can produce W. Choosing an energy or number weight does not by itself prove those ingredients or their propagation in time.

4.2 Operator domains and moments

For normalized states in a common operator domain $D(B_f) \cap D(B_f^*)$, expansion and Cauchy–Schwarz give

$$|\langle B_f\rangle_\psi - \langle B_f\rangle_\chi| \leq \|\psi-\chi\|\big(\|B_f\psi\| + \|B_f^*\chi\|\big). \tag{14}$$

State-distance control plus certified field second moments can therefore suffice. The domains and moments must be valid at the later time of interest.

For a self-adjoint field component O, let $d = \|\rho-\sigma\|_1$ and $C_O = \operatorname{Tr}[(\rho+\sigma)O^2] < \infty$. Spectral truncation at $L > 0$ yields $|\operatorname{Tr}[(\rho-\sigma)O]| \leq Ld + C_O/L$. Optimization gives $2\sqrt{dC_O}$. Applied to both real components of B_f, this produces

$$|\operatorname{Tr}[(\rho-\sigma)B_f]| \leq 2\sqrt{d(C_\times + C_+)}. \tag{15}$$

Any mismatch between the comparison state's expectation and w_h must be added. This route requires actual self-adjoint realizations and finite certified moments.

4.3 Why trace distance alone is insufficient

For an unbounded oscillator quadrature $Q = (a+a^*)/\sqrt{2}$, take vacuum σ and $\rho_\varepsilon = (1-\varepsilon)\sigma + \varepsilon|\alpha\rangle\langle\alpha|$ with real $\alpha = 1/(\sqrt{2}\varepsilon)$. Then $\|\rho_\varepsilon - \sigma\|_1 \leq 2\varepsilon \to 0$, but the quadrature expectation difference is one. Each state has finite moments; no uniform energy bound survives as $\varepsilon \to 0$. Bounded-observable closeness therefore cannot substitute for the required field control.

The supplied continuum also needs its stated form interpretation. Under a literal $d\omega$ smearing, $|u_R(\omega)|^2 \propto \omega j_2(\omega R)^2$ has a logarithmically divergent unweighted ultraviolet norm, although its $1/\omega$-weighted norm is finite. A standard operator estimate requiring $u_R \in L^2$ cannot be imported without further justification. A model-specific form realization, regularization with a complete tail argument, or another domain construction may resolve that issue; this observation is not a claim that the model is undefined.

5. Accuracy targets from retained native inputs

A bounded independent calculation consumes the retained native $E(t)$ upper endpoints, conservative lower endpoints of the Ω and M boxes, and $G_{\mathrm{up}} = \max(|g_{\mathrm{lo}}|, |g_{\mathrm{hi}}|)$. It evaluates Eq. (11) with exact rational arithmetic at $R, 2R, 64, t_q, t_c$, where

$$t_q = 573049\pi/4, \qquad t_c = 573049\pi. \tag{16}$$

All 30 amplitude-margin tests and all 30 budget-positivity tests pass. The table gives a common sufficient target for each lane: the minimum of its five budgets, each rounded downward to 24 decimal places. A valid error certificate strictly below the tabulated value would pass the sufficient test at all five specified endpoints. It would not by itself cover the times between them.

Table 1: Downward-rounded correlator-error targets in the normalized chart. These are allowed error budgets, not achieved bounds on $|\delta w|$.

Lane	Common lower budget
01	0.018288994592828160418234
02	0.008824267054453970901251
03	0.000708877992379492000797
12	0.017051412789380609734264
13	0.000737462089444887158295
23	0.000770798838747397371241

The isolated checker exits zero and passes 120 independent integer identity/rounding checks. Its exact reduced fractions and consumed native cells are retained in the target-budget artifact identified in the accompanying source note. The native source graph and interval values are unchanged; this is not a new GEN6 correlator calculation. In particular, none of these positive budgets establishes W itself.

6. Verification and present scope

Exact symbolic checks verify the reference cancellation, Eq. (4), the budget simplification and saturation of the disk estimate. The completed native evidence remains six positive initial clocks, 30 positive mean-rate bounds, and 15 positive versus 15 zero-containing later pointwise bounds. This note establishes an accuracy target for the missing correlator calculation, not satisfaction of that target.

The next native scientific result must include a valid field-domain/reference construction, the achieved W or residual bound, its continuum and numerical error payments, and a strictly positive common-parameter margin. Any unresolved entries must remain unresolved.

Source basis

The exact source equation and preparation conventions are those of the supplied *Original Write Histories and a Coupled Source–Field Action*, v1.1, and *From Original Write Incidence to Six Received Clocks*, 15 October 2026. Current endpoint status comes from H002628's recovered native records. The centered theorem and sufficient weighted-control conditions above are independent mathematical derivations under explicit hypotheses.